Linear Algebra II HW 8

1. Let $A=\begin{pmatrix}\alpha+\beta & -\alpha\beta \\ 1 & 0\end{pmatrix}$ where α and β are distinct real numbers. Let I be the 2×2 identity matrix.

   (a) Show that $A^2=(\alpha+\beta)A-\alpha\beta I$.

   (b) Using (a), or otherwise, show that $(A-\alpha I)^2=(\beta-\alpha)(A-\alpha I)$ and $(A-\beta I)^2=(\alpha-\beta)(A-\beta I)$.

   (c) Let $X=s(A-\alpha I)$ and $Y=t(A-\beta I)$ where s and t are real numbers.
   Suppose $A=X+Y$.

   (i) Find s and t in terms of α and β .

   (ii) For any positive integer n , prove that
   $$X^n=\frac{\beta^n}{\beta-\alpha}(A-\alpha I) \quad \text{and} \quad Y^n=\frac{\alpha^n}{\alpha-\beta}(A-\beta I) .$$

   (iii) For any positive integer n , express A^n in the form of $pA+qI$, where p and q are real numbers.
   [Note: It is known that for any 2×2 matrices H and K ,
   if $HK=KH=\begin{pmatrix}0 & 0\\ 0 & 0\end{pmatrix}$, then $(H+K)^n=H^n+K^n$ for any positive integer n .]

Solution

1. (a) $$A^2=\begin{pmatrix}\alpha+\beta & -\alpha\beta \\ 1 & 0\end{pmatrix}\begin{pmatrix}\alpha+\beta & -\alpha\beta \\ 1 & 0\end{pmatrix}$$
$$=\begin{pmatrix}(\alpha+\beta)^2-\alpha\beta & -\alpha\beta(\alpha+\beta) \\ \alpha+\beta & -\alpha\beta\end{pmatrix}$$
$$(\alpha+\beta)A-\alpha\beta I=(\alpha+\beta)\begin{pmatrix}\alpha+\beta & -\alpha\beta \\ 1 & 0\end{pmatrix}-\alpha\beta\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$$
$$=\begin{pmatrix}(\alpha+\beta)^2-\alpha\beta & -\alpha\beta(\alpha+\beta) \\ \alpha+\beta & -\alpha\beta\end{pmatrix}$$
i.e. $A^2=(\alpha+\beta)A-\alpha\beta I$

(b) $$(A-\alpha I)^2=A^2-2\alpha A+\alpha^2 I$$
$$=(\alpha+\beta)A-\alpha\beta I-2\alpha A+\alpha^2 I \quad \text{by (a)}$$
$$=(\beta-\alpha)A+(\alpha^2-\alpha\beta)I$$
$$=(\beta-\alpha)(A-\alpha I)$$

By interchanging α and β, we have $(A-\beta I)^2=(\alpha-\beta)(A-\beta I)$.

(c) (i) $A=X+Y$

$$\begin{pmatrix}\alpha+\beta & -\alpha\beta\\ 1 & 0\end{pmatrix}=s\begin{pmatrix}\beta & -\alpha\beta\\ 1 & -\alpha\end{pmatrix}+t\begin{pmatrix}\alpha & -\alpha\beta\\ 1 & -\beta\end{pmatrix}$$

$$=\begin{pmatrix}s\beta+t\alpha & -\alpha\beta(s+t)\\ s+t & -s\alpha-t\beta\end{pmatrix}$$

Comparing the entries, we have $\begin{cases}s\beta+t\alpha=\alpha+\beta\\ s+t=1\\ s\alpha+t\beta=0\end{cases}$

Solving, $s=\dfrac{\beta}{\beta-\alpha}$ and $t=\dfrac{\alpha}{\alpha-\beta}$

(ii) Consider the statement " $X^n=\dfrac{\beta^n}{\beta-\alpha}(A-\alpha I)$ and $Y^n=\dfrac{\alpha^n}{\alpha-\beta}(A-\beta I)$ ".

When $n=1$, $X=\dfrac{\beta}{\beta-\alpha}(A-\alpha I)$ and $Y=\dfrac{\alpha}{\alpha-\beta}(A-\beta I)$ are true by (c)(i)

Assume $X^k=\dfrac{\beta^k}{\beta-\alpha}(A-\alpha I)$ and $Y^k=\dfrac{\alpha^k}{\alpha-\beta}(A-\beta I)$, where k is a positive integer.

$$X^{k+1}=\frac{\beta^k}{\beta-\alpha}(A-\alpha I)\frac{\beta}{\beta-\alpha}(A-\alpha I) \quad \text{by the assumption}$$

$$=\frac{\beta^{k+1}}{(\beta-\alpha)^2}(\beta-\alpha)(A-\alpha I) \quad \text{by (b)}$$

$$=\frac{\beta^{k+1}}{\beta-\alpha}(A-\alpha I)$$

$$Y^{k+1}=\frac{\alpha^k}{\alpha-\beta}(A-\beta I)\frac{\alpha}{\alpha-\beta}(A-\beta I) \quad \text{by the assumption}$$

$$=\frac{\alpha^{k+1}}{(\alpha-\beta)^2}(\alpha-\beta)(A-\beta I) \quad \text{by (b)}$$

$$=\frac{\alpha^{k+1}}{\alpha-\beta}(A-\beta I)$$

Hence the statement is true for $n = k+1$.
By the principle of mathematical induction, the statement is true for all positive integers n .

(iii) $XY = s(A-\alpha I)t(A-\beta I)$

$= st[A^2-(\alpha+\beta)A+\alpha\beta I]$ by (a)

$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$YX = t(A-\beta I)s(A-\alpha I)$

$= st[A^2-(\alpha+\beta)A+\alpha\beta I]$ by (a)

$= \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$A^n = (X+Y)^n$

$= X^n + Y^n$ by the note given

$= \dfrac{\beta^n}{\beta-\alpha}(A-\alpha I)+\dfrac{\alpha^n}{\alpha-\beta}(A-\beta I)$ by (ii)

$= \dfrac{\alpha^n-\beta^n}{\alpha-\beta}A+\dfrac{\alpha\beta^n-\alpha^n\beta}{\alpha-\beta}I$

2. Find an orthogonal basis for the column space of $A = \begin{pmatrix} 1 & 3 & 5 \\ 1 & 1 & 0 \\ 1 & 1 & 2 \\ 1 & 3 & 3 \end{pmatrix}$. Find a *QR* factorization of *A*. Hence find the least-squares solution of the inconsistent system $A\boldsymbol{x} = \boldsymbol{b}$ where $\boldsymbol{b} = (1,2,3,4)^T$.

Solution

$\boldsymbol{v}_1 = \boldsymbol{x}_1$

$\boldsymbol{v}_2 = \boldsymbol{x}_2 - \dfrac{\boldsymbol{x}_2 \cdot \boldsymbol{v}_1}{\boldsymbol{v}_1 \cdot \boldsymbol{v}_1}\boldsymbol{v}_1 = \boldsymbol{x}_2 - 2\boldsymbol{v}_1 = (1,-1,-1,1)^T$

$\boldsymbol{v}_3 = \boldsymbol{x}_3 - \dfrac{\boldsymbol{x}_3 \cdot \boldsymbol{v}_1}{\boldsymbol{v}_1 \cdot \boldsymbol{v}_1}\boldsymbol{v}_1 - \dfrac{\boldsymbol{x}_3 \cdot \boldsymbol{v}_2}{\boldsymbol{v}_2 \cdot \boldsymbol{v}_2}\boldsymbol{v}_2 = \boldsymbol{x}_3 - 2.5\boldsymbol{v}_1 - 1.5\boldsymbol{v}_2 = (1,-1,1,-1)^T$

$$Q = \begin{pmatrix} 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 \\ 1/2 & -1/2 & 1/2 \\ 1/2 & 1/2 & -1/2 \end{pmatrix}, \quad R = \begin{pmatrix} 2 & 4 & 5 \\ 0 & 2 & 3 \\ 0 & 0 & 2 \end{pmatrix}$$

The least-squares solution is $R^{-1}Q\boldsymbol{b} = \begin{pmatrix} 1/2 & -1 & 1/4 \\ 0 & 1/2 & -3/4 \\ 0 & 0 & 1/2 \end{pmatrix}\begin{pmatrix} 1/2 & 1/2 & 1/2 & 1/2 \\ 1/2 & -1/2 & -1/2 & 1/2 \\ 1/2 & -1/2 & 1/2 & -1/2 \end{pmatrix}\begin{pmatrix} 1 \\ 2 \\ 3 \\ 4 \end{pmatrix} = \begin{pmatrix} 2.25 \\ 0.75 \\ -0.5 \end{pmatrix}$

Find the eigenvalues and eigenvectors of the matrices.

3. $A = \begin{pmatrix} 3 & -1 \\ 4 & -2 \end{pmatrix}$.

Solution

The eigenvalues λ and eigenvectors $\mathbf{x}$ satisfy the equation $(\mathbf{A} - \lambda\mathbf{I})\mathbf{x} = \mathbf{0}$, or

$$\begin{pmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{31}$$

The eigenvalues are the roots of the equation

$$\det(\mathbf{A} - \lambda\mathbf{I}) = \begin{vmatrix} 3-\lambda & -1 \\ 4 & -2-\lambda \end{vmatrix} = \lambda^2 - \lambda - 2 = 0. \tag{32}$$

Thus the eigenvalues are $\lambda_1 = 2$ and $\lambda_2 = -1$.

To find the eigenvectors we return to Eq. (31) and replace λ by each of the eigenvalues in turn. For $\lambda = 2$ we have

$$\begin{pmatrix} 1 & -1 \\ 4 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}. \tag{33}$$

Hence each row of this vector equation leads to the condition $x_1 - x_2 = 0$, so x_1 and x_2 are equal, but their value is not determined. If $x_1 = c$, then $x_2 = c$ also and the eigenvector $\mathbf{x}^{(1)}$ is

$$\mathbf{x}^{(1)} = c \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad c \neq 0. \tag{34}$$

Usually, we will drop the arbitrary constant c when finding eigenvectors; thus instead of Eq. (34) we write

$$\mathbf{x}^{(1)} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{35}$$

and remember that any nonzero multiple of this vector is also an eigenvector. We say that $\mathbf{x}^{(1)}$ is the eigenvector corresponding to the eigenvalue $\lambda_1 = 2$.

4. $A = \begin{pmatrix} 1 & 0 \\ 6 & -1 \end{pmatrix}$.

Solution

Since A is triangular, its eigenvalues are obviously ± 1.

For $\lambda = 1$: $A - 1I = \begin{bmatrix} 0 & 0 \\ 6 & -2 \end{bmatrix}$. The equation $(A - 1I)\mathbf{x} = \mathbf{0}$ amounts to $6x_1 - 2x_2 = 0$, so $x_1 = (1/3)x_2$ with x_2 free. The general solution is $x_2 \begin{bmatrix} 1/3 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \end{bmatrix}$.

For $\lambda = -1$: $A + 1I = \begin{bmatrix} 2 & 0 \\ 6 & 0 \end{bmatrix}$. The equation $(A + 1I)\mathbf{x} = \mathbf{0}$ amounts to $2x_1 = 0$, so $x_1 = 0$ with x_2 free. The general solution is $x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}$, and a basis vector for the eigenspace is $\mathbf{v}_2 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$.

From $\mathbf{v}_1$ and $\mathbf{v}_2$ construct $P = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}$. Then set $D = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, where the eigenvalues in D correspond to $\mathbf{v}_1$ and $\mathbf{v}_2$ respectively.

5. $A = \begin{pmatrix} -1 & 4 & -2 \\ -3 & 4 & 0 \\ -3 & 1 & 3 \end{pmatrix}$.

Solution

The eigenvalues of A are given to be 1, 2, and 3.

For $\lambda = 3$: $A - 3I = \begin{bmatrix} -4 & 4 & -2 \\ -3 & 1 & 0 \\ -3 & 1 & 0 \end{bmatrix}$, and row reducing $\begin{bmatrix} A - 3I & \mathbf{0} \end{bmatrix}$ yields $\begin{bmatrix} 1 & 0 & -1/4 & 0 \\ 0 & 1 & -3/4 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general solution is $x_3 \begin{bmatrix} 1/4 \\ 3/4 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 3 \\ 4 \end{bmatrix}$.

For $\lambda = 2$: $A - 2I = \begin{bmatrix} -3 & 4 & -2 \\ -3 & 2 & 0 \\ -3 & 1 & 1 \end{bmatrix}$, and row reducing $\begin{bmatrix} A - 2I & \mathbf{0} \end{bmatrix}$ yields $\begin{bmatrix} 1 & 0 & -2/3 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general solution is $x_3 \begin{bmatrix} 2/3 \\ 1 \\ 1 \end{bmatrix}$, and a nice basis vector for the eigenspace is $\mathbf{v}_2 = \begin{bmatrix} 2 \\ 3 \\ 3 \end{bmatrix}$.

For $\lambda = 1$: $A - I = \begin{bmatrix} -2 & 4 & -2 \\ -3 & 3 & 0 \\ -3 & 1 & 2 \end{bmatrix}$, and row reducing $\begin{bmatrix} A - 1I & \mathbf{0} \end{bmatrix}$ yields $\begin{bmatrix} 1 & 0 & -1 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. The general solution is $x_3 \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$, and a basis vector for the eigenspace is $\mathbf{v}_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$.

From $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ construct $P = \begin{bmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 1 \\ 3 & 3 & 1 \\ 4 & 3 & 1 \end{bmatrix}$. Then set $D = \begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$, where the eigenvalues in D correspond to $\mathbf{v}_1, \mathbf{v}_2$ and $\mathbf{v}_3$ respectively.